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M.Sc. in High Performance Computing

Final project: Lasso and sparse compressed sensing

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1 Introduction

Compressed Sensing (CS) theory breaks through the limitations of the Nyquist sampling theorem, indicating that if a signal is sparse in a certain transform domain, it can be precisely reconstructed using far less data than conventionally required. This theory holds extremely high application value in fields such as medical imaging (e.g., rapid MRI scanning), radar detection, and image compression.

In compressed sensing, the reconstruction of a sparse signal is typically formulated as a convex optimization problem regularized by the ℓ_1 norm, known as the Lasso (Least Absolute Shrinkage and Selection Operator) problem. Because the ℓ_1 norm is non-differentiable at the origin, traditional gradient descent methods cannot be directly applied to solve it. Therefore, a series of efficient non-smooth optimization algorithms have emerged in the fields of machine learning and numerical analysis.

The core objective of this project is to independently implement four major numerical optimization algorithms: ISTA, FISTA, CD, and ADMM. Furthermore, these algorithms are applied to the reconstruction tasks of 1D signals and 2D medical images. Through a quantitative comparison of multiple dimensions (goodness of fit, MSE, runtime, convergence speed), we explore the performance boundaries of each algorithm in practical applications.

2 Algorithm

In compressed sensing, the observation data $y \in \mathbb{R}^M$ and the unknown signal $x \in \mathbb{R}^N$ $(M < N)$ satisfy the linear relationship $y = Ax + \epsilon$, where A is the measurement matrix and ϵ is Gaussian noise. Given the sparsity of x, we reconstruct it by minimizing the following Lasso objective function:

$$\min_x f(x) = \frac{1}{2}\|Ax - y\|_2^2 + \lambda\|x\|_1 \tag{1}$$

Due to the non-differentiability of the ℓ_1 norm, we introduce the Soft Thresholding Operator as its Proximal Operator:

$$S_\kappa(v) = \operatorname{sgn}(v)\max(|v| - \kappa, 0) \tag{2}$$

2.1 Iterative Shrinkage-Thresholding Algorithm (ISTA)

ISTA is a specific instance of the proximal gradient descent method applied to the Lasso problem. In each iteration step, gradient descent is first performed along the differentiable quadratic loss component, and then the soft thresholding operator is applied to handle the ℓ_1 regularization term:

$$x_{k+1} = S_{\alpha\lambda}\left(x_k - \alpha A^T(Ax_k - y)\right) \tag{3}$$

where $\alpha \leq 1/L$ is the step size, and $L = \|A\|_2^2$ is the Lipschitz constant of the smooth part of the objective function.

2.2 Fast Iterative Shrinkage-Thresholding Algorithm (FISTA)

To address the slow convergence speed of ISTA (convergence rate $O(1/k)$), FISTA introduces the Nesterov momentum acceleration mechanism, improving its convergence rate to $O(1/k^2)$. The iterative format is as follows:

$$x_{k+1} = S_{\alpha\lambda}\left(z_k - \alpha A^T(Az_k - y)\right) \tag{4}$$

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2} \tag{5}$$

$$z_{k+1} = x_{k+1} + \frac{t_k - 1}{t_{k+1}}(x_{k+1} - x_k) \tag{6}$$

2.3 Coordinate Descent (CD)

Coordinate descent minimizes the objective with respect to only one component of the variable x, denoted as x_j, at a time, while fixing the other components. For the Lasso problem, each coordinate subproblem has an analytical solution, which avoids the sensitivity to the step size in global gradient calculations. The formula is:

$$x_j \leftarrow \frac{S_\lambda(A_j^T r + \|A_j\|_2^2 x_j)}{\|A_j\|_2^2} \tag{7}$$

where $r = y - Ax$ is the current residual.

2.4 Alternating Direction Method of Multipliers (ADMM)

ADMM decomposes the original problem into independently solvable subproblems by introducing an auxiliary variable z. The constrained problem is written as: $\min \frac{1}{2}\|Ax - y\|_2^2 + \lambda\|z\|_1$, s.t. $x - z = 0$. Using the Augmented Lagrangian method, the update formulas are derived as:

$$x_{k+1} = (A^T A + \rho I)^{-1}(A^T y + \rho(z_k - u_k)) \tag{8}$$

$$z_{k+1} = S_{\lambda/\rho}(x_{k+1} + u_k) \tag{9}$$

$$u_{k+1} = u_k + x_{k+1} - z_{k+1} \tag{10}$$

3 Experiment 1: 1D Sparse Signal Recovery

3.1 Experimental Setup

We generated a ground truth signal x_{true} with a dimension of $N = 500$, a measurement dimension of $M = 150$, and a sparsity of $S = 10$. The measurement matrix A adopts a Gaussian random matrix, and Gaussian noise with standard deviation $\sigma = 0.05$ is added to the observations y. Four algorithms were used to reconstruct it.

3.2 Results and Analysis

Experimental results indicate that under identical initial conditions:

- **Convergence Speed**: The objective function of ADMM drops extremely fast, converging in fewer than 60 steps. Utilizing momentum acceleration, FISTA converges in approximately 150 steps. ISTA is the slowest, requiring nearly 200 steps. While CD involves fewer iterations in terms of Epochs, each Epoch inherently contains N internal loops.

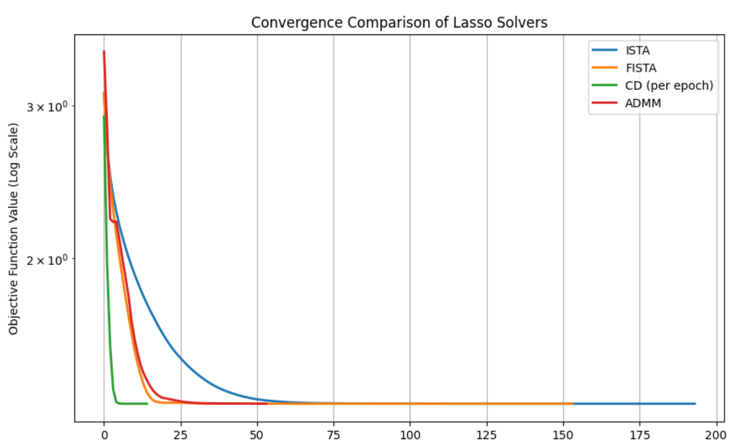

Figure 1: Convergence comparison of lasso solvers

- **Recovery Accuracy**: All four algorithms accurately pinpointed the support set of non-zero elements of the signal, and the reconstructed signals were highly consistent with the original signal at all peaks.

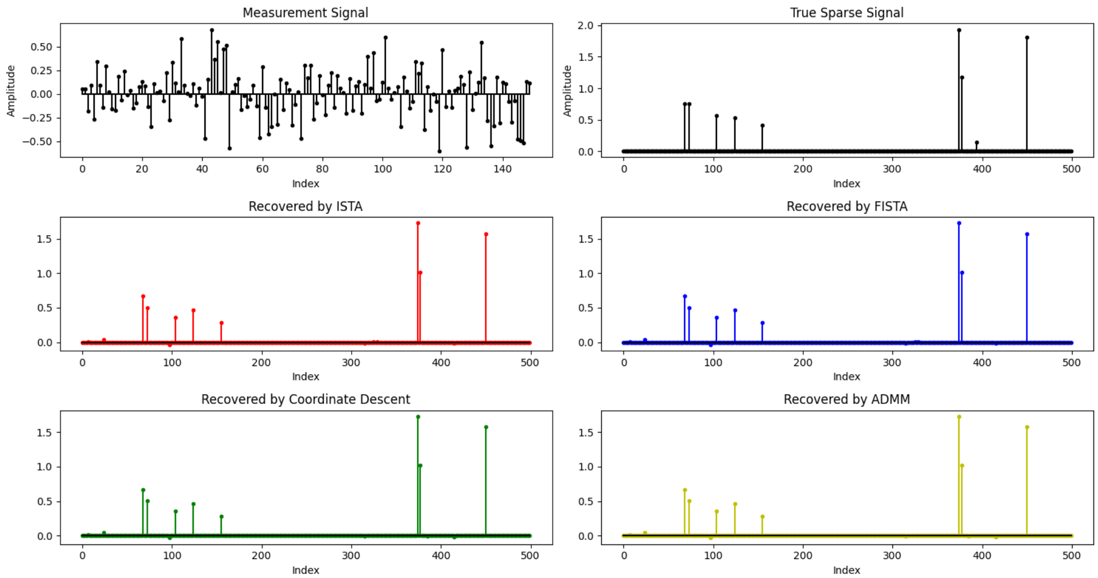

Figure 2: Signal recovery using different solvers

Solver	**MSE**	R^2	**Iteration**	**Runtime(s)**
ISTA	0.000571	0.9720	193	0.093
FISTA	0.000571	0.9720	153	0.058
CD	0.000570	0.9720	14	0.200
ADMM	0.000570	0.9720	53	0.167

Table 1: Solver Performance

4 Experiment 2: 2D Medical Image Compressed Reconstruction (MRI Simulation)

4.1 Variable Density Fourier Sampling and Sparsification Strategy

In real Magnetic Resonance Imaging (MRI) applications, applying purely uniform random sampling directly to high-dimensional images results in extremely poor reconstruction quality. Therefore, we introduced the following critical improvement strategies:

1. **Sparsifying Basis Selection**: The 2D Inverse Discrete Cosine Transform (IDCT) matrix was used as the transform basis D to project the non-sparse Shepp-Logan phantom from the spatial domain to the frequency domain, rendering it highly sparse.

2. **Variable Density Sampling**: The vast majority of MRI image energy is concentrated in the low-frequency region. When generating the sampling mask, we **enforced 100% full sampling in the ultra-low frequency central region (radius** ≤ 6), while for the peripheral high-frequency

regions, the sampling probability decreased proportionally to the inverse cube of the distance. The overall sampling rate M/N for the experiment was set to 60%.

3. **Complex-to-Real Decoupling**: The matrix generated by Fourier sampling is complex-valued. To reuse our implemented real-domain Lasso solvers, the complex equations were decomposed and concatenated into real and imaginary parts: $A_{real} = [\Re(A)^T, \Im(A)^T]^T$.

4.2 Evaluation Metrics and Comparative Results

We executed the reconstruction process using the four algorithms on an image of size 32×32, with a maximum of 2000 iterations and a tolerance of 10^{-5}. The evaluation metrics were Mean Squared Error (MSE) and Goodness of Fit (R^2).

The variable density sampling scheme produced extremely clear reconstructed images, effectively suppressing artifacts. The R^2 scores obtained by all four algorithms successfully surpassed the 0.98 threshold, proving the exceptional feasibility of the proposed scheme in compressed sensing image reconstruction.

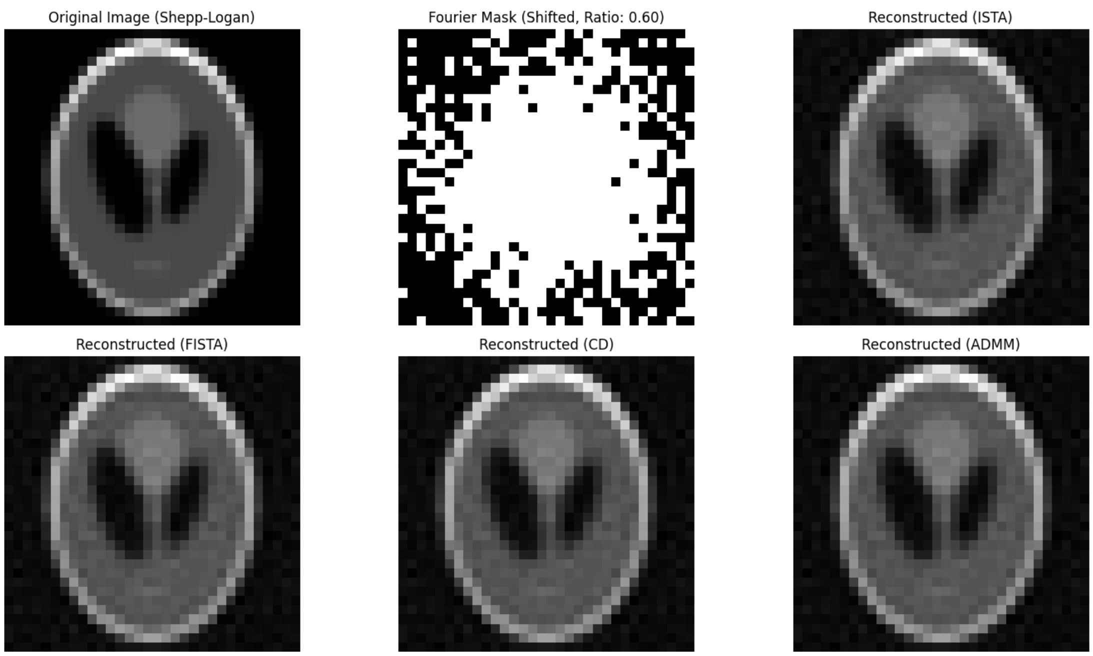

Figure 3: Image reconstruction using different solvers

Solver	**MSE**	R^2	**Iteration**	**Runtime(s)**
ISTA	0.000317	0.9853	229	4.803
FISTA	0.000317	0.9853	140	1.450
CD	0.000317	0.9853	59	6.357
ADMM	0.000317	0.9853	229	1.392

Table 2: Solver Performance

5 Conclusion

Combining the charts and data from the image and signal reconstructions, this study draws the following conclusions:

1. **ADMM**: It holds an overwhelming advantage in terms of the "number of iterations," converging exceptionally fast. However, its update step involves computing the inverse of the matrix $A^T A + \rho I$. When the signal dimension is large like experiment 2, even with pre-computation and caching of the inverse matrix, the constant overhead of a single matrix multiplication remains enormous, resulting in suboptimal overall iterations.

2. **CD (Coordinate Descent)**: Because it cannot perform global parallel matrix-vector multiplications, executing large-scale for-loops in an interpreted language like Python causes a severe performance bottleneck. Consequently, its computation time far exceeds the other algorithms.

3. **ISTA**: It is the simplest to implement, requires no matrix inversion, and has extremely fast single steps. However, limited by the sublinear convergence rate of first-order gradients, it necessitates a large number of global iteration steps.

4. **FISTA**: **Demonstrates the best overall performance.** Its single-step overhead is identical to that of ISTA (involving only matrix-vector multiplications), but leveraging Nesterov momentum drastically reduces the number of iterations. It showcases the most outstanding engineering practicality in trading off "computational time" against "reconstruction quality," making it the preferred solution for handling such large-scale ℓ_1 optimization problems.

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